SALES FORECASTING & DEMAND INTELLIGENCE SYSTEM

Executive Business Report

Project: Internship Project – Week 3 & Week 4

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Data Period: 2015-01-03 to 2018-12-30

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EXECUTIVE SUMMARY

This project analysed four years of retail sales data (9,800 transactions across 3 product categories and 4 regions, generating $2.26 million in revenue) to build an end-to-end Sales Forecasting and Demand Intelligence System. Three forecasting models were evaluated, with SARIMA achieving the best performance (MAPE: 20.53%), making it the most accurate model for this dataset. The analysis also identified 62 unusual sales days, likely linked to promotions or bulk corporate purchases, and segmented 17 product sub-categories into demand-based groups to support inventory planning. The complete solution was deployed as an interactive Streamlit dashboard for exploring sales trends, forecasts, anomalies, and product demand patterns.

KEY FINDINGS

• Technology generated the highest revenue: approximately $0.83 million (36.6% of total sales).

• Sales show strong seasonality, with November and December consistently recording the highest sales.

• Average shipping time across all regions was 4.0 days.

• A total of 4,922 customer orders were processed with an average order value of approximately $459.

3-MONTH SALES FORECAST

Using the SARIMA model, average monthly sales are forecast at approximately $82,986, with expected monthly sales ranging from $60,332 to $97,168. The model achieved an MAE of $19,244.49 and a MAPE of 20.53%, providing a reliable basis for inventory planning.

Model Comparison

SARIMA → MAE $19,244.49 RMSE $19,950.07 MAPE 20.53%

Prophet → MAE $20,296.01 RMSE $22,487.47 MAPE 21.89%

XGBoost → MAE $28,035.82 RMSE $28,063.84 MAPE 31.18%

Recommended production model: SARIMA (lowest error across all evaluation metrics).

TOP 3 UNUSUAL SALES DAYS

1. 2015-03-18 — $28,107 (likely a large bulk B2B or corporate purchase).

2. 2017-10-02 — $18,453 (likely a large corporate order or promotional event).

3. 2018-10-22 — $15,159 (likely a high-value customer purchase or special sales campaign).

Isolation Forest and Z-Score anomaly detection achieved a 50.0% agreement rate, providing additional confidence that these events represent genuine business anomalies.

PRODUCT DEMAND SEGMENTATION

Seventeen product sub-categories were grouped using K-Means clustering.

• Stars (5): High sales and stable demand. Maintain high inventory and continued marketing.

• Question Marks (3): High-value but volatile products. Monitor closely, maintain safety stock, and investigate demand fluctuations.

• Dogs (9): Lower-value, volatile products. Review profitability, minimise inventory investment, and consider discontinuation where appropriate.

• Cash Cows: No product sub-categories met the criteria during the analysis period.

BUSINESS RECOMMENDATIONS

1. Increase inventory allocation for Furniture. Furniture demonstrated the strongest overall growth (+149.0%), indicating increasing demand and a higher risk of stock shortages.

2. Prepare inventory for the November–December peak season. Begin procurement from September to reduce stockout risk during the busiest sales period.

3. Implement automated daily anomaly monitoring. The 62 detected high-sales anomalies suggest opportunities to identify successful promotions and investigate unusual sales events more quickly.

RISK / LIMITATION

The forecasting system assumes future demand will follow historical patterns. While SARIMA captures recurring trends and seasonality effectively, it cannot predict unexpected events such as supply chain disruptions, economic downturns, regulatory changes, new competitors, or major shifts in customer preferences. The model should be retrained regularly with new data and used as a decision-support tool alongside business expertise.

CONCLUSION

The proposed Sales Forecasting and Demand Intelligence System provides a practical framework for improving inventory planning, forecasting future demand, identifying unusual sales behaviour, and supporting data-driven business decisions. By combining forecasting, anomaly detection, product segmentation, and an interactive Streamlit dashboard, the solution enables managers to make more informed operational and inventory decisions.

Report based on analysis of 9,800 transactions conducted between 2015-01-03 and 2018-12-30.

Supporting materials:

• analysis.ipynb

• Streamlit Dashboard

• Forecasting, anomaly detection, and product segmentation results